ROCSCIENCE

Case4_with_surcharge
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case4_with_surcharge

Project Summary

File Name:	Case4_with_surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:11.428s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	30
Circles per division:	15
Number of iterations:	15
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	50
Orientation:	Vertical

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.159740
Center:	20.009, 70.632
Radius:	46.713
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	59.061, 45.000
Resisting Moment:	68237.4 kN-m
Driving Moment:	58838.6 kN-m
Total Slice Area:	122.42 m2
Surface Horizontal Width:	29.0611 m
Surface Average Height:	4.2125 m

Method: spencer

FS	1.155970
Center:	19.963, 70.604
Radius:	46.695
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	59.012, 45.000
Resisting Moment:	67600 kN-m
Driving Moment:	58479.1 kN-m
Resisting Horizontal Force:	1165.24 kN
Driving Horizontal Force:	1008.02 kN
Total Slice Area:	121.68 m2
Surface Horizontal Width:	29.0125 m
Surface Average Height:	4.19407 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	33898
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	29376
Number of Invalid Surfaces:	4522

Error Codes

Error Code -108 reported for 741 surfaces
Error Code -111 reported for 3781 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.15974

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.16244	6.13528	13.0862	Material 1	9	32	9.42381	10.9292	3.08732	0	3.08732	5.27792	5.27792
2	1.16244	18.1128	14.5507	Material 1	9	32	14.1736	16.4376	11.9027	0	11.9027	15.5816	15.5816
3	1.16244	29.4974	16.0291	Material 1	9	32	18.5597	21.5244	20.0433	0	20.0433	25.3754	25.3754
4	1.16244	40.2758	17.5184	Material 1	9	32	22.587	26.1951	27.5179	0	27.5179	34.6475	34.6475
5	1.16244	50.4333	19.0201	Material 1	9	32	26.2593	30.454	34.3335	0	34.3335	43.3856	43.3856
6	1.16244	59.9537	20.5355	Material 1	9	32	29.5794	34.3044	40.4955	0	40.4955	51.5756	51.5756
7	1.16244	68.8185	22.066	Material 1	9	32	32.5491	37.7485	46.0073	0	46.0073	59.2016	59.2016
8	1.16244	77.0069	23.6134	Material 1	9	32	35.1696	40.7876	50.8707	0	50.8707	66.2457	66.2457
9	1.16244	84.4952	25.1792	Material 1	9	32	37.4407	43.4215	55.0859	0	55.0859	72.6876	72.6876
10	1.16244	91.2569	26.7655	Material 1	9	32	39.3617	45.6493	58.6512	0	58.6512	78.5044	78.5044
11	1.16244	97.262	28.3742	Material 1	9	32	40.9306	47.4689	61.5631	0	61.5631	83.6704	83.6704
12	1.16244	102.476	30.0078	Material 1	9	32	42.1446	48.8768	63.8162	0	63.8162	88.156	88.156
13	1.16244	106.861	31.6687	Material 1	9	32	42.9997	49.8685	65.4032	0	65.4032	91.928	91.928
14	1.16244	110.372	33.3599	Material 1	9	32	43.4909	50.4381	66.3148	0	66.3148	94.9481	94.9481
15	1.16244	112.958	35.0847	Material 1	9	32	43.6118	50.5783	66.5392	0	66.5392	97.1727	97.1727
16	1.16244	114.56	36.8468	Material 1	9	32	43.3548	50.2803	66.0622	0	66.0622	98.551	98.551
17	1.16244	115.11	38.6506	Material 1	9	32	42.7108	49.5334	64.867	0	64.867	99.0244	99.0244
18	1.16244	114.528	40.5011	Material 1	9	32	41.669	48.3252	62.9335	0	62.9335	98.5235	98.5235
19	1.16244	112.719	42.4042	Material 1	9	32	40.2169	46.6412	60.2385	0	60.2385	96.967	96.967
20	1.16244	109.569	44.3669	Material 1	9	32	38.3401	44.4645	56.7551	0	56.7551	94.2571	94.2571
21	1.16244	104.94	46.398	Material 1	9	32	36.0214	41.7755	52.4517	0	52.4517	90.2753	90.2753
22	1.16244	96.5565	48.5079	Material 1	9	32	40.8977	47.4307	61.5017	0	61.5017	107.741	107.741
23	1.16244	73.3404	50.7099	Material 1	9	32	41.419	48.0353	62.4694	0	62.4694	113.091	113.091
24	1.16244	45.7718	53.021	Material 1	9	32	32.5934	37.7999	46.0895	0	46.0895	89.3754	89.3754
25	1.16244	15.7076	55.4638	Material 1	9	32	23.5465	27.3078	29.2986	0	29.2986	63.5126	63.5126

Global Minimum Query (spencer) - Safety Factor: 1.15597

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.1605	6.1024	13.1486	Material 1	9	32	12.9222	14.9377	9.50222	0	9.50222	12.5209	12.5209
2	1.1605	18.0154	14.6109	Material 1	9	32	18.4594	21.3385	19.7458	0	19.7458	24.5579	24.5579
3	1.1605	29.338	16.0877	Material 1	9	32	23.3305	26.9694	28.757	0	28.757	35.4857	35.4857
4	1.1605	40.0565	17.5756	Material 1	9	32	27.5847	31.8871	36.627	0	36.627	45.3644	45.3644
5	1.1605	50.1566	19.0758	Material 1	9	32	31.2642	36.1405	43.4339	0	43.4339	54.2453	54.2453
6	1.1605	59.6218	20.5898	Material 1	9	32	34.4063	39.7726	49.2465	0	49.2465	62.172	62.172
7	1.1605	68.4337	22.1189	Material 1	9	32	37.0434	42.8211	54.1251	0	54.1251	69.1812	69.1812
8	1.1605	76.5716	23.6649	Material 1	9	32	39.2044	45.3191	58.1228	0	58.1228	75.3036	75.3036
9	1.1605	84.0118	25.2293	Material 1	9	32	40.9143	47.2957	61.2859	0	61.2859	80.5643	80.5643
10	1.1605	90.728	26.8142	Material 1	9	32	42.1955	48.7767	63.6561	0	63.6561	84.9836	84.9836
11	1.1605	96.69	28.4215	Material 1	9	32	43.0676	49.7849	65.2695	0	65.2695	88.577	88.577
12	1.1605	101.864	30.0537	Material 1	9	32	43.5481	50.3403	66.1584	0	66.1584	91.3554	91.3554
13	1.1605	106.211	31.7133	Material 1	9	32	43.6524	50.4609	66.3513	0	66.3513	93.3255	93.3255
14	1.1605	109.687	33.4031	Material 1	9	32	43.3941	50.1623	65.8734	0	65.8734	94.49	94.49
15	1.1605	112.241	35.1266	Material 1	9	32	42.7852	49.4584	64.747	0	64.747	94.8466	94.8466
16	1.1605	113.814	36.8873	Material 1	9	32	41.8365	48.3617	62.9918	0	62.9918	94.3891	94.3891
17	1.1605	114.338	38.6897	Material 1	9	32	40.5572	46.8829	60.6254	0	60.6254	93.1059	93.1059
18	1.1605	113.734	40.5388	Material 1	9	32	38.9559	45.0319	57.6632	0	57.6632	90.9804	90.9804
19	1.1605	111.905	42.4405	Material 1	9	32	37.0402	42.8174	54.1191	0	54.1191	87.9895	87.9895
20	1.1605	108.74	44.4019	Material 1	9	32	34.8167	40.2471	50.0058	0	50.0058	84.1032	84.1032
21	1.1605	104.101	46.4316	Material 1	9	32	32.292	37.3286	45.3353	0	45.3353	79.2827	79.2827
22	1.1605	96.0139	48.5401	Material 1	9	32	35.1975	40.6873	50.7102	0	50.7102	90.5499	90.5499
23	1.1605	73.1728	50.7406	Material 1	9	32	35.4648	40.9962	51.2046	0	51.2046	94.5967	94.5967
24	1.1605	45.6667	53.0503	Material 1	9	32	27.994	32.3602	37.3842	0	37.3842	74.6013	74.6013
25	1.1605	15.6714	55.4916	Material 1	9	32	20.6813	23.907	23.8562	0	23.8562	53.9383	53.9383

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.15974

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30	25	0	0	0
2	31.1624	25.2702	10.1204	0	0
3	32.3249	25.5719	23.005	0	0
4	33.4873	25.9059	37.8859	0	0
5	34.6498	26.2728	54.0451	0	0
6	35.8122	26.6735	70.812	0	0
7	36.9747	27.109	87.563	0	0
8	38.1371	27.5802	103.72	0	0
9	39.2995	28.0884	118.751	0	0
10	40.462	28.6349	132.17	0	0
11	41.6244	29.2212	143.538	0	0
12	42.7869	29.849	152.465	0	0
13	43.9493	30.5204	158.613	0	0
14	45.1117	31.2375	161.699	0	0
15	46.2742	32.0028	161.503	0	0
16	47.4366	32.8193	157.869	0	0
17	48.5991	33.6904	150.72	0	0
18	49.7615	34.62	140.065	0	0
19	50.924	35.6129	126.019	0	0
20	52.0864	36.6745	108.819	0	0
21	53.2488	37.8115	88.8547	0	0
22	54.4113	39.0321	66.705	0	0
23	55.5737	40.3464	33.4163	0	0
24	56.7362	41.7671	-7.18849	0	0
25	57.8986	43.3109	-40.4529	0	0
26	59.0611	45	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.15597

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30	25	0	0	0
2	31.1605	25.2711	12.42	8.42827	34.161
3	32.321	25.5736	27.8684	18.9116	34.161
4	33.4815	25.9083	45.3185	30.7532	34.1609
5	34.642	26.2759	63.8665	43.34	34.1609
6	35.8025	26.6772	82.7178	56.1325	34.1609
7	36.963	27.1132	101.176	68.6583	34.1609
8	38.1235	27.5849	118.635	80.5059	34.1609
9	39.284	28.0934	134.571	91.3205	34.161
10	40.4445	28.6402	148.54	100.8	34.161
11	41.605	29.2268	160.169	108.691	34.1609
12	42.7655	29.8549	169.156	114.79	34.161
13	43.926	30.5263	175.27	118.939	34.161
14	45.0865	31.2434	178.347	121.027	34.161
15	46.247	32.0087	178.292	120.99	34.161
16	47.4075	32.8251	175.084	118.812	34.1608
17	48.568	33.6961	168.773	114.53	34.161
18	49.7285	34.6255	159.495	108.233	34.1607
19	50.889	35.618	147.471	100.074	34.1609
20	52.0495	36.6792	133.025	90.271	34.1609
21	53.21	37.8157	116.597	79.1227	34.1608
22	54.3705	39.0357	98.7626	67.0205	34.1609
23	55.531	40.3492	72.9982	49.5367	34.1609
24	56.6915	41.7691	41.449	28.1274	34.1609
25	57.852	43.312	16.2576	11.0325	34.161
26	59.0125	45	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Distributed Load	55, 45 95, 45	Constant DistributionOrientation: VerticalMagnitude: 50 kN/m2Creates Excess Pore Pressure: No